

SESSION 2: BUILDING AN AIR-POWERED ARM

The Power of Your Breath

Did you know that your breath can move things? Today, we aren't just blowing out candles—we are going to use the air from our lungs to power a mechanical robot arm!

| What is Pneumatics?

It means using air to make things move! Just like a balloon gets bigger when you blow into it, that same pushing power can move machines and tools.

| Invisible but Strong

Air is all around us. Even though we can't see it, it is very strong! When we trap air inside a tube or a balloon and squeeze it, it creates a "push" that can lift heavy objects.

Pneumatics vs. Hydraulics

In our last session, we used water to lift things. Today, we use air. Let's see the difference!

Pneumatics	Hydraulics
Uses Air	Uses Liquid
Light and Fast	Strong and Heavy
Safe and Clean!	Can be messy (Oil/Water)

The "Squish" Factor

Air is "squishy" (compressible), which makes it great for movements that need to be fast and springy. This is why kids can safely play with air-powered toys!

Why Use Air Power?

Pneumatics are used everywhere because they are easy to control and work with simple parts.

Indian Context: The "Hiss" of the Bus

Have you ever stood near a big Volvo or local luxury bus and heard a loud "Hissss" sound when the doors open? That is compressed air escaping! The doors are moved by pneumatic cylinders.

- **Cycle Pumps:** When you pump air into your bicycle tire at the local repair shop, you are using pneumatic energy.
- **Air Brakes:** Heavy trucks on Indian highways use air brakes to stop safely.
- **Festive Horns:** The loud horns used during cricket matches or festivals use air to create sound.

Your Engineering Kit

To build our robot arm, we need these simple materials:

- Cardboard (For the arm structure)
- Balloons (Our air muscles!)
- Tubing (The air highway)
- Straws
- Tape & Glue
- Ruler & Markers
- Scissors
- Air Bladders (Optional)

Safety First: Always use scissors under an adult's supervision!

| How to Build Your Arm

Follow these steps to bring your mechanical arm to life:

Step 1: Cut Parts

Cut two long strips of cardboard. These will be the "bones" of your arm.

Step 2: Connect the Joint

Use a pin or a straw to join the two cardboard strips so they can pivot (move) like an elbow.

Step 3: Attach the Muscle

Tape a balloon between the two strips of cardboard. Connect your plastic tubing to the mouth of the balloon and tape it tight!

Step 4: Connect the Air Source

Attach the other end of the tube to a syringe or just prepare to use your own breath!

Test Your Invention!

Now comes the fun part—making it move!

The Science: When you blow air into the tube, the balloon inflates (gets bigger). As it grows, it pushes against the cardboard parts. This makes the arm move up and down!

The "Air Punch" Challenge

Can your pneumatic arm do work? Let's find out!

1. Set up a tower of three paper cups.
2. Position your air-powered arm next to the tower.
3. Take a deep breath and blow into the tube!
4. Did you knock the tower over? Yes / No

Engineer's Note: If the arm is too slow, try blowing faster!

Fun Pneumatics Quiz

Answer these questions to see if you are a Pneumatic Pilot!

1. **What does Pneumatics use to move things?**

Answer: Air!

2. **Is air power faster or slower than liquid power?**

Answer: Faster!

3. **Can you find pneumatics on a local Indian bus?**

Answer: Yes, in the doors!

Conclusion

Today, you used the invisible power of air to create movement. Whether it's lifting a robot arm or stopping a massive truck, air power is an amazing tool for engineers. Keep exploring!

Certified Planrs Academy Pneumatic Pilot!